3D CoSim: Coupled Operator Learning-Based Co-Simulator for Transferable 3D-IC Analysis

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2025 IEEE/ACM International Conference On Computer Aided Design (ICCAD) | 979-8-3315-1560-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICCAD66269.2025.11240711

Abstract*—The rapid evolution of 3D integrated circuits (ICs) has ushered in unprecedented integration density and performance gains. However, this advancement intensifies the challenges of multi-physics interactions, particularly the intricate coupling between electromagnetic (EM) and thermal fields. Thermal and EM optimizations of 3D ICs often require extensive and complex Partial Differential Equation (PDE) simulations. Neural network-based field solvers are renowned for their high efficiency and accuracy. However, existing approaches are often trained using results from traditional physical field solvers, which are time-consuming and lack compatibility with more complex design configurations. In this paper, *for the first time*, we propose 3D CoSim, a Physics-Informed Neural Network (PINN) based simulator for multi-physics analysis with arbitrary 3D IC configurations. 3D CoSim has a sophisticated computational framework that learns the nonlinear functional mappings from geometrical and physical setups to multiphysics fields. Leveraging a Multi-input DeepONet architecture, 3D CoSim integrates multiple PDE configurations. This integration encompasses the heat equation, Maxwell's equations, and a diverse range of boundary conditions, thereby enabling the derivation of a unified solution for both EM and thermal fields. This integration delivers a paradigm shift in simulation efficiency. Experimental results demonstrate that 3D CoSim achieves a speedup of 844× to 7600× compared to traditional COMSOL while maintaining comparable accuracy of 97% for EM simulation and 99.8% for thermal simulation.

***Index Terms*—3D IC, multiphysics simulator, PINN, operator learning**

I. Introduction

3D integrated circuit (IC) technology, enabled by advanced heterogeneous integration and vertical stacking architectures [1, 2], has become a cornerstone of integrated circuit development in the post-Moore era. 3D ICs are stacked with multiple active silicon layers, which are interconnected by silicon through vias (TSVs), resulting in their higher power density [3]. By leveraging 3D interconnection paradigms, 3D ICs have demonstrated transformative potential in high-performance computing, AI accelerators, and 5G/6G RF systems [4–6], However, the high-density interconnects and heterogeneous material stacking inherent to 3D integration exacerbate several critical challenges, including electromagnetic (EM) crosstalk, thermal stress concentration, and multiphysics coupling. These issues demand an innovative co-design simulator to identify errors and optimize architectural performance while ensuring reliability and efficiency.

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Partial differential equation (PDE) solvers play a fundamental role in simulating and analyzing physical fields in different domains [7]. Traditional field solvers, such as the Finite Element Method (FEM), Finite Difference Method (FDM), and Finite Volume Method (FVM), provide numerical solutions to complex multiphysics problems by discretizing the governing equations over a computational domain and these methods form the foundation of widely used commercial tools such as COMSOL Multiphysics and ANSYS [8, 9]. Nevertheless, as the integration density of 3D ICs increases, traditional field solvers face significant computational challenges, particularly in handling memory-intensive, high-fidelity models.

To address these limitations, numerous field solvers based on neural networks have been proposed, and the computational efficiency has been greatly improved [10–12]. However, existing approaches still suffer from critical limitations: (1) a heavy reliance on traditional field solvers for generating training data [10]. (2) difficulty in learning continuous function space mappings, leading to poor generalization to unknown design parameters [11–13]. (3) the absence of effective schemes for predicting vector fields with both amplitude and phase components, which hinders accurate solutions for frequency-domain three-dimensional EM fields; (4) a lack of unified multiphysics co-simulation frameworks capable of capturing the bidirectional coupling between EM and thermal effects in 3D ICs. Physics-Informed Neural Networks (PINNs) offer a transformative alternative to traditional field solvers by enabling PDE-constrained training without relying on extensive data. PINNs directly encode governing equations, *e.g.*, Maxwell's equations and Heating Equations, into the loss function through automatic differentiation.

Co-simulation is a multiphysics field coupling analysis technique that aims to accurately capture the nonlinear interaction effects in complex systems by solving the governing equations of multiple interacting physical fields, *e.g.*, EM and thermal, in a synchronized way [14–17]. In 3D IC design, the coupling between EM and thermal fields is particularly significant. Joule heating induced by EM fields alters the local electrical conductivity, while temperature variations, in turn, influence EM field distributions, forming a dynamic, bidirectional feedback loop. However, existing work usually adopts a sequential iterative strategy, which requires repeated calls to independent solvers for data transfer. This strategy is computationally expensive and prone to error accumulation due to the differences in temporal/spatial scales between fields. While PINNs exhibit inherent advantages in single-physics modeling through PDE-constrained learning, their application to bidirectional field interactions faces critical challenges of convergence failure and training instability when enforcing shared latent space constraints. Therefore, it is essential to develop a new co-simulation paradigm that enables high-precision and high-efficiency predictions.

The contributions of this work are highlighted below:

- *For the first time*, a multiphysics-coupled operator learning-based co-simulator for arbitrary 3D IC is proposed to realize EM-Thermal analysis based on PDE under different physical domains.
- Under multi-input DeepONet with vector field neural network, 3D CoSim is proposed to implement transferable EM field extraction of arbitrary 3D ICs in the frequency domain.
- The innovative framework of 3D CoSim couples shared neural networks for joint simulation of multi-physics fields, avoiding the iterative error accumulation of traditional sequential solving.
- The proposed 3D CoSim achieves 844× to 7600× speed up compared to commercial software COMSOL while maintaining a comparable accuracy of 97% for EM simulation and 99.8% for thermal simulation.

II. Preliminaries: Governing Equations and Boundary Conditions

EM-Thermal co-simulation mainly depends on the combination of steady-state EM analysis and thermal analysis. To enable the co-simulation of 3D ICs, the governing PDEs, along with the corresponding boundary conditions (BCs) on the boundary $\partial\Omega$, must be specified within the domain Ω, which is assumed to be Lipschitz continuous and piecewise smooth. The two configurations are presented as follows. For simplicity, isotropic materials are considered in this work, and material property parameters, *e.g.*, k, ε, μ, are equal to a real value in all directions. The geometry of the 3D IC is modeled as a series of stacked rectangular blocks, with EM and thermal field configurations defined for each layer.

A. Thermal Field Configuration

By solving the heat equation globally, the thermal simulation predicts the temperature distribution within a given chip. The heat equation concerning temperature T is:

$$\nabla \cdot (k(T)\nabla T) + q_V = \rho c_p \frac{\partial T}{\partial t} \qquad in\ \Omega, \tag{1}$$

where $k(T)$ is the temperature-dependent thermal conductivity, q_V is the rate of internal energy production per unit volume at any spatial and temporal location, ρ is the material density, and c_p is the heat capacity. To implement the co-simulation, the steady-state heat equation by setting $\frac{\partial T}{\partial t} = 0$ in 3D ICs with Joule heating as the dominant source term is as follows:

$$\nabla \cdot (k(T)\nabla T) + Q = 0 \qquad in\ \Omega, \tag{2}$$

where $Q = q_V + Q_{EM} = q_V + \sigma(T)\,|E|^2$, $\sigma(T)$ is temperature-dependent electric conductivity, and E represents the electric fields.

Given the flow and dissipation of heat at the boundary, the following types of BCs are considered.

For **Neumann** boundaries, on the package surface with heat flux q_n, heat flux is proportional to the temperature gradient:

$$-k(T)\nabla T \cdot \hat{n} = q_n \qquad on\ \partial\Omega, \tag{3}$$

For **Robin** boundaries, like air-cooled surfaces, the gradient of the temperature is proportional to the difference between the boundary temperature and the ambient temperature $T_{ambient} = 298.15\ K$:

$$-k(T)\nabla T \cdot \hat{n} = h(T - T_{ambient}) \qquad on\ \partial\Omega, \tag{4}$$

where h refers to the convective heat transfer coefficient.

B. EM Field Configuration

Maxwell's equations are usually utilized to depict the basic behavior of the EM field. Herein, to simulate the EM field in 3D ICs in the frequency domain, complex time-harmonic Maxwell's equations are considered as follows:

$$\begin{cases} \nabla \times E = jw\mu(T)H \\ \nabla \times H = -jw\varepsilon(T)E \end{cases} \qquad in\ \Omega, \tag{5}$$

where ω is the angular frequency of the EM field, and H refers to the magnetic field. In addition, the permittivity ε and the permeability μ depend on the properties of the material in the chip, which is also related to the temperature.

To better adapt the PINN models, the Helmholtz equation is deduced from the complex Maxwell's equations (5):

$$\nabla \times \mu^{-1}\nabla \times E - \kappa^2 \varepsilon E = 0 \qquad in\ \Omega, \tag{6}$$

where $\kappa = \omega\sqrt{\varepsilon_0\mu_0}$ is the wave number.

Contrary to the approximate BCs proposed by traditional field solvers, the neural network-based 3D CoSim employs more intrinsic BCs based on physical principles. For time-harmonic EM fields, the BCs are as follows:

$$\mathfrak{B}[u(\mathbf{x})] = 0 \qquad x \in \partial\Omega. \tag{7}$$

For the configurations of PDEs $u(\mathbf{x})$, $\mathfrak{B}$ is the general form of BC operators, which can take various forms depending on the type of boundary. For perfect electric conductors, such as TSV metal-filled layers and copper interconnects, the boundary condition imposes that the tangential component of the electric field vanishes, i.e., $\hat{n} \times E = 0$ on $\partial\Omega$. At the excitation port, the EM field is specified by an imposed field containing complex voltage and phase information: $E = E_{\text{Port}}$ on $\partial\Omega$. To suppress spurious reflections in the simulation of 3D packages, a perfectly matched layer is introduced, enforcing the condition $\hat{n} \cdot E = 0$ on $\partial\Omega$.

The design configurations above represent multi-physics field setups of the PDE for 3D ICs. By parameterizing the structure, 3D CoSim predicts multi-physics fields through operator learning.

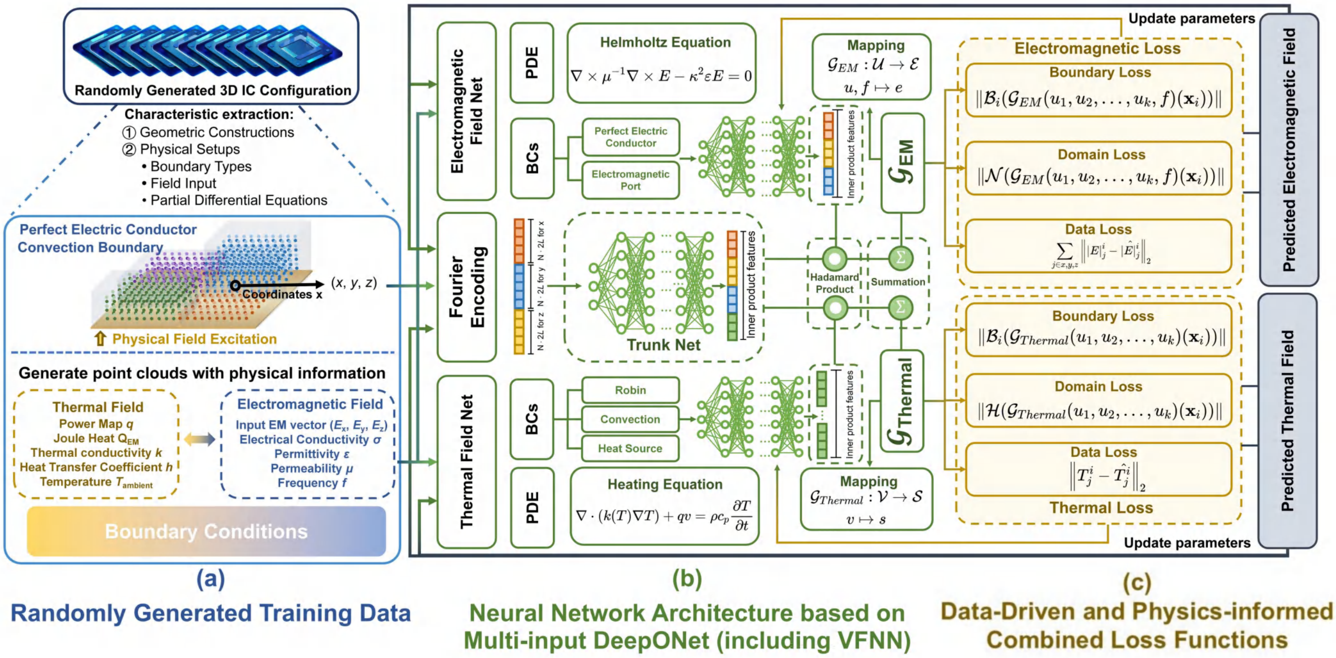

Fig. 1: The Proposed 3D CoSim Framework: (a) Randomly Generated Training Data; (b) Fourier Encoding and 3D CoSim Architecture Design; (c) Loss Functions for Combined Data-Driven Pre-Training and Physics-Informed Post-Training.

III. The 3D CoSim Framework

In this section, we present the 3D CoSim framework for multi-physics simulation. Based on physics-informed DeepONet, 3D CoSim enables the prediction of EM and thermal fields in one neural network architecture. For various design configurations, DeepONet primarily consists of two components, the trunk net and the branch net. The trunk net mainly comprises the positions of the output functions by encoding the coordinates and physical setups, and the branch net consists mostly of the construction of the associated physical fields. The proposed 3D CoSim framework is shown in Fig. 1.

A. Coordinate-wise Fourier Feature Encoding

Deep networks are suitable for learning low-frequency functions; however, 3D CoSim requires fine modeling for the 3D IC with high frequency [18]. Mapping the input coordinates to a higher-dimensional space enables better fitting of data containing high-frequency variations. Herein, Fourier-based encoding is employed to map coordinate elements $\mathbf{x} = (x, y, z)$ on a 3D IC from $\mathbb{R}$ to a higher $2L$-dimensional space $\mathbb{R}^{2L}$. This encoding introduces periodic basis functions at multiple frequencies, enabling the network to more accurately approximate oscillatory solutions. By mitigating the spectral bias of neural networks toward low-frequency components, it enhances the model's ability to capture sharp transitions and fine-scale spatial variations, which are essential for modeling high-frequency EM fields and localized heat sources. Formally, the encoding function is:

$$\gamma(p) = (\sin(2^0\pi p), \cos(2^0\pi p), \cdots, \sin(2^{L-1}\pi p), \cos(2^{L-1}\pi p)). \quad (8)$$

This function $\gamma(\cdot) \in \mathbb{R}^{2L}$ is individually applied to three coordinate points in x, which are normalized to $[0, 1]$. In our experiment, we set $L = 6$ for $\gamma(\mathbf{x})$ to the three components of the Cartesian.

B. Multi-input DeepONet via Hadamard Product

Generally, the dominant equations of different modules are denoted in the following format:

$$\begin{aligned} \mathcal{H}(s(v_1, v_2, \ldots, v_k)(\mathbf{x})) = 0, \\ \mathcal{N}(\mathbf{e}(u_1, u_2, \ldots, u_k, f)(\mathbf{x})) = 0. \end{aligned} \quad (9)$$

Herein, $\mathcal{H}$ and $\mathcal{N}$ are symbolic representations of governing equations (2) and (6), respectively. The solutions of dominant equations for relevant modules are represented as s and $\mathbf{e}$, describing the distribution of temperature and EM fields. A specific distribution field is determined by a particular 3D IC design, which is closely associated with various PDE configurations. To achieve transferable learning of the entire 3D IC, for different functional modules, different sets of PDE configurations ($\mathcal{U} : \{u_i; f\}_{i=1}^k \in \mathbb{C}^{k+1}$ for the EM module and $\mathcal{V} : \{v_i\}_{i=1}^k \in \mathbb{R}^k$ for the Heat module) are considered. By incorporating the corresponding coordinates, the prediction for the target function space $\mathcal{S}$ and $\mathcal{E}$ can be obtained as the solutions to the governing equations.

Multi-input DeepONet employs neural networks to learn the solution operators, mapping input functions to PDE solutions. The regression for the solution operators $\mathcal{G}_{EM}$ and $\mathcal{G}_{Thermal}$ are formulated as:

$$\begin{aligned} \mathcal{G}_{EM} : \mathcal{U} \to \mathcal{E} \qquad & u, f \mapsto \mathbf{e}, \\ \mathcal{G}_{Thermal} : \mathcal{V} \to \mathcal{S} \qquad & v \mapsto s. \end{aligned} \quad (10)$$

The DeepONet mainly consists of two components: the trunk net f for learning the coordinate information $\mathbf{x} \in \mathbb{R}$ and outputs the basis functions; and the branch net g, which takes the discretized PDE configurations $\mathcal{U}$ and $\mathcal{V}$ as input. Then the approximate network is:

$$\tilde{\mathcal{G}}(v_1, \cdots, v_n)(\mathbf{x}) = \mathcal{S}\left(\tilde{g}_1(v_1) \odot \cdots \odot \tilde{g}_n(v_n) \odot \tilde{f}(\mathbf{x})\right) + b. \quad (11)$$

where $\odot$ is the Hadamard product, $\mathcal{S}$ is the summation of all components in the vector, and b is a trainable bias term. Building such operator mapping, 3D CoSim can capture features from PDE configurations to predict transferable physical fields $\mathcal{S}$ and $\mathcal{E}$.

C. Frequency-adaptive VFNN for EM Module

In the EM simulation module, given the properties of the amplitude and phase of the EM field, the branch net is proposed to construct operator mappings between complex inputs $\mathcal{U} : \{u_i; f\}_{i=1}^{k} \in \mathbb{C}^{k+1}$ and outputs $\Gamma_{EM} \in \mathbb{C}^{q}$. For simplicity, the EM field inputs have the same phase, so only the magnitudes are considered for the input electric field, which can be restricted to the set of real numbers $\mathbb{R}^{k+1}$. This design aligns with the harmonic nature of Maxwell's equations, eliminates information loss caused by decoupling real and imaginary parts, and avoids employing complex neural networks. The operator mapping relationship is as follows:

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_k \end{bmatrix} \rightarrow \begin{bmatrix} \Re(\gamma_1) & \Im(\gamma_1) \\ \Re(\gamma_2) & \Im(\gamma_2) \\ \vdots & \vdots \\ \Re(\gamma_k) & \Im(\gamma_k) \end{bmatrix}. \quad (12)$$

Meanwhile, considering the vector nature of the EM field, vector field neural networks (VFNN) are utilized. Thus, 3D CoSim can realize the transition from scalar to vector field prediction. The implementation principle is depicted in Fig. 2. By segmenting the output, the VFNN enables directional decomposition of the electric field, allowing accurate characterization along each axis. Next, the outputs of the two sub-networks are processed, and the operator mapping of the component electric fields in all directions can be calculated. The operators in each direction are jointly trained under the physical constraints.

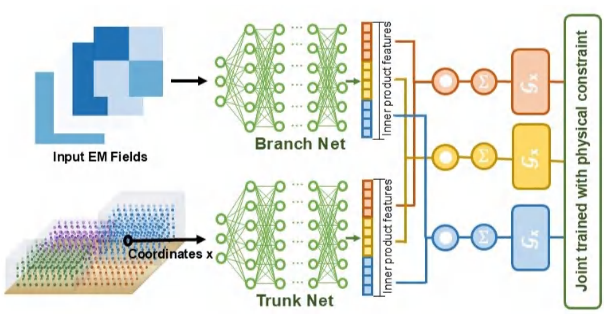

Fig. 2: Workflow of Vector Field Neural Networks.

D. Encoding the Characteristic for 2/3 D Input

Combining the VFNN with DeepONet, 3D CoSim realizes the prediction under different PDE configurations. In general, we consider k PDE configurations and correspondingly k different input functions. And for i^{th} configurations, random samples from PDE configurations $\mathcal{U}$ and $\mathcal{V}$ are identified on certain specific coordinates, *e.g.*, boundary, port, and input as a vector into the i^{th} branch nets. By repeating this process across all k design configurations, 3D CoSim can sample and encode both the geometric and physical attributes of a given design.. A well-defined input order of physical fields is essential for reliable coupled-field simulation using 3D CoSim. Consider the geometric configuration illustrated on the left side of Fig.1. For this quad-core chip, we define the input functions of thermal and EM fields at its bottom: a two-dimensional electric field vector map as input, which allows for arbitrary placement of electric field excitations between cores, as well as a two-dimensional power map, which supports arbitrary placement of the heat source. These input functions are identified and recognized, and stored as physical information in the point cloud. Simultaneously, we flatten these inputs into a vector and feed them into the Branch Nets. Additionally, Joule heating induced by EM fields contributes as an internal heat source in the thermal domain. The 3D power (q_V) map is defined by the Heating equation in Eq. (2), and the remaining procedures follow the same approach. In each iteration, the EM field is predicted first, and its output is subsequently used as input for the thermal module. The loss functions for the two physical fields are computed and passed back to the corresponding branch net and trunk net to update the parameters and train the network for co-simulation.

E. Training 3D CoSim with Physics-informed Loss

DeepONet is generally trained by obtaining a large amount of data based on a traditional field solver and does not act as a simulator. In chip verification, for a single FEM simulation, it takes a lot of time to realize. Consequently, training and deploying data-driven neural networks for chip-level physical field prediction remain computationally expensive. Thus, 3D CoSim follows the innovations of PHYSICAL INFORMATION to solve PDEs, thus training multiple mappings of actual physical implications. Training different patterns within each epoch to learn a global mapping between arbitrary patterns and EM (thermal) field distributions is necessary. However, we observe experimentally that the prediction results based entirely on loss is inaccurate, sometimes even leading to non-convergence between the coupled fields. Therefore, we first pre-train 3D CoSim based on a few data points obtained from the field solver, and subsequently further train it via the physical constraint loss. Inspired by PINN, the loss function of 3D CoSim is denoted as follows:

$$\mathcal{L} = \sum_{i}^{Domain} (\mathcal{L}_i^{Boundary} + \mathcal{L}_i^{Inner}). \quad (13)$$

As illustrated in Eq. (13), 3D CoSim partitions the entire IC domain into boundary and interior points, which are evaluated independently.

For the Inner point, the loss is mainly determined by the PDEs: the heating equation for thermal simulation and the Helmholtz equation for EM simulation. For a chip structure with k points, we first index all the coordinates located in its configuration, denoted as y_i, and compute the loss $\mathcal{L}_i^{Inner}$ in parallel as follows:

$$\begin{aligned}\mathcal{L}_i^{Inner} &= \left\|\mathcal{N}(\mathcal{G}_{EM}(u_1, u_2, \ldots, u_k, f)(\mathbf{x}_i))\right\|, \\ \mathcal{L}_i^{Inner} &= \left\|\mathcal{H}(\mathcal{G}_{Thermal}(u_1, u_2, \ldots, u_k)(\mathbf{x}_i))\right\|.\end{aligned} \tag{14}$$

The entire simulation domain is regularized using the governing PDEs as physical constraints. For boundary points i, depending on different BCs, including Neumann, Convection boundaries, and so on, the physical mapping will be constrained by the following loss function $\mathcal{L}_i^{Boundary}$:

$$\begin{aligned}\mathcal{L}_i^{Boundary} &= \left\|\mathcal{B}_i(\mathcal{G}_{EM}(u_1, u_2, \ldots, u_k, f)(\mathbf{x}_i))\right\|, \\ \mathcal{L}_i^{Boundary} &= \left\|\mathcal{B}_i(\mathcal{G}_{Thermal}(u_1, u_2, \ldots, u_k)(\mathbf{x}_i))\right\|,\end{aligned} \tag{15}$$

where $\mathcal{B}_i$ denotes the equation corresponding to the BC at the i^{th} point in the thermal/EM field simulation. For simplicity, boundary conditions (BCs) with identical properties are assigned to all points across different physical domains. In addition, gradient descent is based on an automatic differentiation algorithm to minimize the total loss. Moreover, gradient descent based on an automatic differentiation algorithm to minimize the total loss is employed to train 3D CoSim.

IV. Experiments

In this section, we present the 3D CoSim simulation implementation under different physical field couplings. Furthermore, we compare the performance of our proposed method with that of the commercial software COMSOL Multiphysics. Our results illustrate that for any construction, the trained 3D CoSim achieves 7600x speedup while maintaining high accuracy of 97% for EM simulation and 99.8% for thermal simulation.

A. Data Preparation and Experiment Setup

TABLE I: Geometric configurations of 3D ICs for testing.

	Chip 1	Chip 2	Chip 3
Layer size (mm)	$1 \times 1 \times 0.1$	$1 \times 1 \times 0.1$	$0.5 \times 1 \times 0.1$
Number of Layers	non-uniform	uniform	uniform
TSV	diameter: 0.01, pitch: 0.01		
Heat transfer coefficient	25 W/(m2·K)		

Generally, we choose three different 3D ICs for EM-Thermal co-simulation evaluation, and the geometric configuration is presented in Table I. 3D CoSim equivalent of the heat sink is designed on top of the chip as a convection boundary. Chips 1 and 2 are quad-core 3D-stacked architectures, and the core is the Alpha 21364 microprocessor. To evaluate the compatibility of 3D CoSim with varying hardware architectures, Chips 1 and 2 are constructed with distinct stacking configurations by varying the number of device layers per core. As shown in Fig. 3, each core of chip 1 is composed of a stack of different layers, with layers 3, 4, 1, and 2, respectively, while chip 2 consists of two device layers. Chip 3 is an octa-core 3D stacked chip architecture with the same 2 device layers of the same size. The cores are connected with TSVs, herein, for simplicity, which can be considered to enable lossless transmission of thermal and EM fields. Physical excitations (EM, thermal) are input at the yellow substrate. The

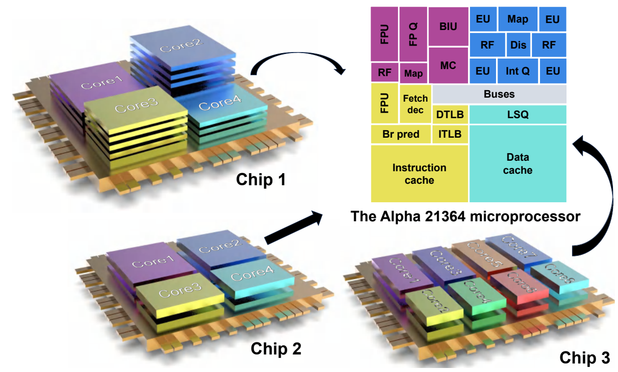

Fig. 3: The schematic of the 3D IC architecture.

input EM signals control the EM field distribution across the 3D IC, so we focus on the performance of the 3D CoSim with unknown EM field inputs. The input EM field can be decomposed into three directions $E = (E_x, E_y, Ez)$, and the intensity of the electric field in the z-direction is 10 times larger than the others. The EM field input to each core is assumed to have identical intensity and phase. Power controls heat generation in 3D ICs, and 3D CoSim considers both 2D and 3D heat sources. 2D power is fed via the substrate, while Joule heat generated by EM fields enables 3D power input to the chip.

Our framework is developed by Pytorch 2.1.2 and Python 3.9. The experimental procedures are conducted on a Linux server with an AMD EPYC 9754 128-core Processor CPU and an NVIDIA A100 80GB GPU.

B. Generate Point Clouds with Physical Information

Based on the geometric configuration of 3D ICs, we consider a $41 \times 41 \times 21$ mesh grid array-based cubic geometry, which contains a 3D IC in an arbitrary stacked configuration inside it. The maximum size of the chip in practice is $2{\times}2{\times}0.4$ (mm). By recognizing the features of randomly generated chips, 3D CoSim accordingly generates point clouds that record their geometrical and physical properties. If not specified, the default simulated chip is assumed to be composed of silicon. The thermal thermal conductivity is $k = 0.1$ W/(m·K), while electric conductivity is $\sigma = 2.5 \times 10^{-4}$ S/m. The relative permittivity and permeability are $\epsilon_r = 11.7$ and $\mu_r = 1$, respectively. In addition, each point stored the location (boundary or domain), the EM field information (vector), the Joule heat, and the temperature corresponding to each point.

C. Dual DeepONet Settings

In our implementation, we employ a dual DeepONet architecture for coupled electro-thermal field prediction, with specialized configurations for both the electric field and thermal field models. We utilize a DeepONet architecture with a 3-layer branch net with 64 neurons per layer, combined with a 3-layer trunk net with 128 neurons per layer. DeepONets are trained independently for thermal and EM fields. For thermal DeepONet, sub-networks produce 50-dimensional feature vectors passing into the inner product feature space. For EM simulation, the vector field DeepONet is applied,

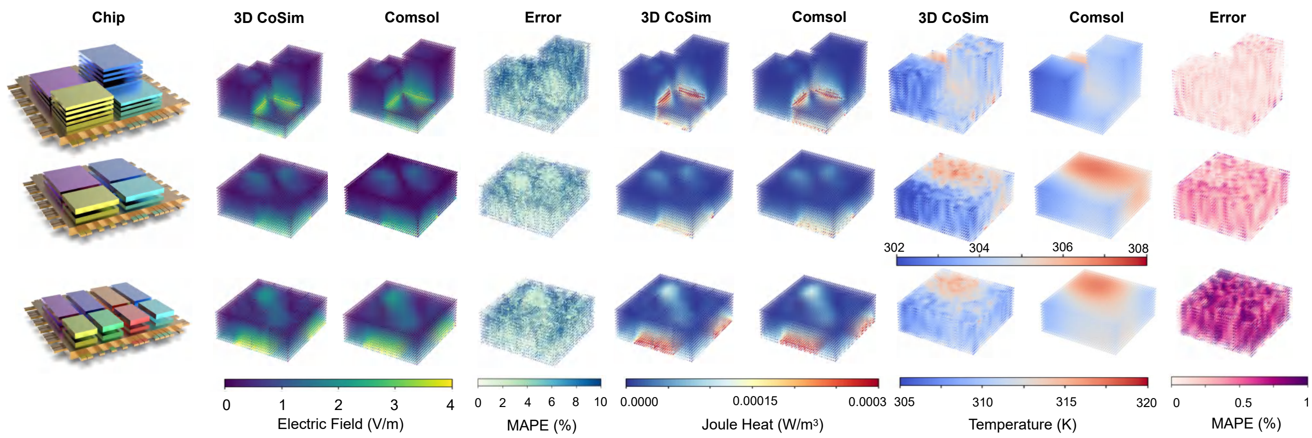

Fig. 4: Predicted multiphysics fields for different geometry configurations.

which essentially respects the vector nature of the EM field. The architecture divides the 150 inner product features into three equal groups and trains mapping relations to predict each field component. The SiLU activation function ensures a smooth gradient across the network, which is essential for accurate prediction of the field near the boundary.

We trained 3D CoSim by 2500 iterations to ensure convergence, taking 15 hours on a signal NVIDIA A100 80GB GPU. In each iteration, all generated points in the chip domain are fed to the trunk net, for example, 19049 points in Chip 1. The randomly generated excitation function (power, EM field) is fed to the branch net, and each core has the same excitation. Through experiments, it is found that 3D CoSim based on PDEs and BCs faces the problem of poor accuracy and non-convergence between thermal and EM fields. To address this, we adopted a two-stage training strategy: an initial data-driven phase followed by PINN-based refinement. This approach facilitates accurate modeling of the coupled-field interactions. The initial learning rate is set to 1e-3, with a decay factor of 0.9 applied every 100 iterations. In the testing phase, for novel chip geometry configurations and excitations, we conduct a pointwise comparison between predicted fields and COMSOL results.

D. Prediction Results with Different Geometric Configurations

To comprehensively demonstrate the accuracy and effectiveness of 3D CoSim, we compared the predicted results with Comsol, which specializes in multiphysics field simulation. The 3D CoSim performance was verified by testing the above three sets of chips with different geometric configurations (4 core: Chip 1&2, 8 core: Chip 3), and their mean absolute percentage errors (MAPEs) are reported in the Table II.

TABLE II: Mean error for different chip configurations.

	Chip 1	Chip 2	Chip 3
Thermal MAPE (%)	0.12	0.31	0.67
EM MAPE (%)	3.71	2.57	2.73

Generally, 3D CoSim has a high accuracy in predicting different geometric configurations. All predictions are shown in Fig. 4, where EM fields, joule heat fields, and temperature fields are well calculated. It is worth emphasizing that, considering that many of the training samples are based on Chip 1 or its quad-core approximate configurations, 3D CoSim demonstrated high accuracy in predicting the temperature field and EM field. However, due to the rapid decay of the EM field intensity in the far field, The weak electric field intensity in the far-field increases numerical error, and thus, the electric field prediction of Chip 1 is not as effective numerically as that of Chips 2 and 3.

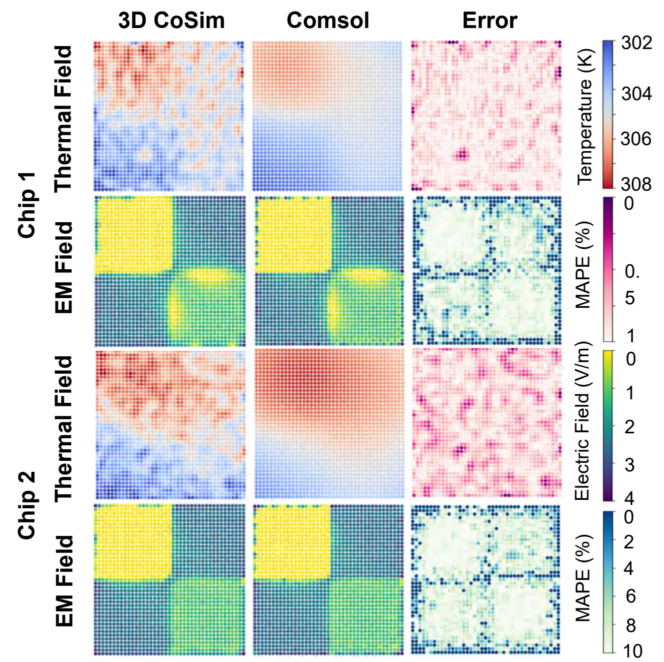

Fig. 5: Physical fields at the predicted excitation port interface (z=0) for different geometrical configurations.

In predicting an octocore chip, the presence of more sophisticated excitation functions at the ports leads to an exponential increase in port power, and the entire chip has a higher temperature as a result. The high prediction accuracy demonstrates the strong generalization capability of 3D CoSim across diverse geometric configurations in both EM and thermal field simulations, and all the trained geometry configurations, which

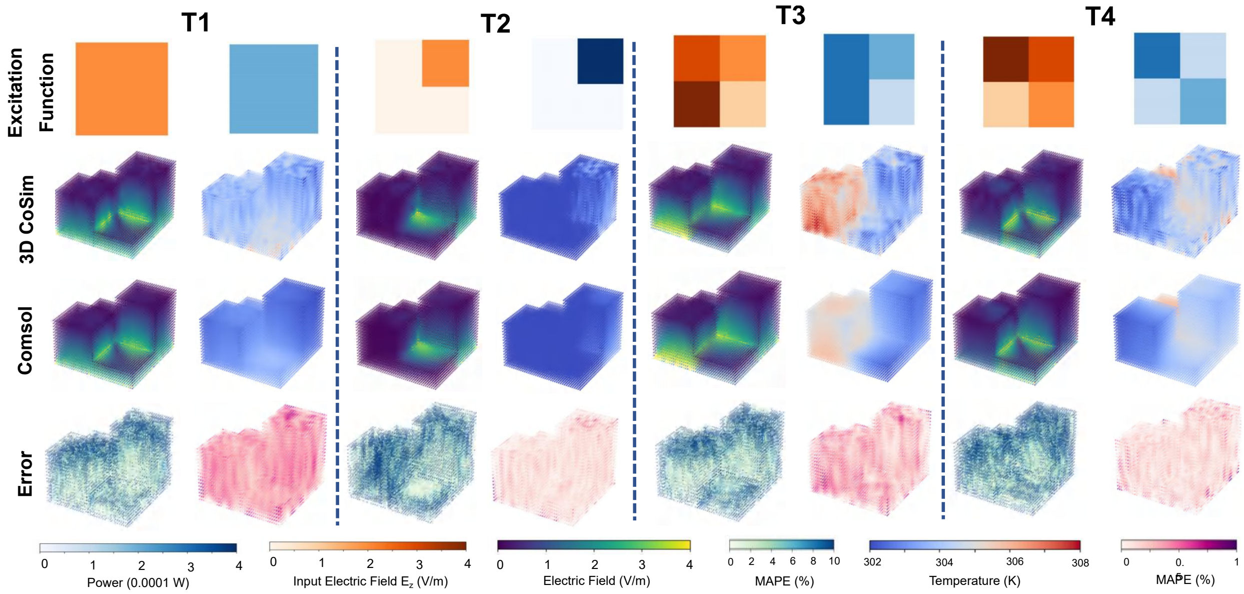

Fig. 6: Predicted multiphysics fields for different excitation functions.

are composed of quad-core chips based on different randomly generated layer stacks. However, accurate predictions are made not only for quad-core but also for octa-core chips, demonstrating that 3D CoSim gives satisfactory predictions for different geometric configurations.

E. Prediction Results with Different Excitation Functions

To evaluate the predictive performance of 3D CoSim under different excitation equations, several groups of test samples are generated. All the results for these cases are shown in Fig. 6. From left to right, there are four groups of test cases, which we refer to as T1, T2, T3, and T4. The prediction performance of 3D CoSim for different test cases is shown in Table III, and all predictions are shown in Fig. 7. The results indicate that 3D CoSim maintains high predictive accuracy at the excitation port under a variety of input functions. 3D CoSim manages points on the boundary well and realizes high-accuracy prediction.

TABLE III: Mean error for different excitation functions.

	T1	T2	T3	T4
Thermal MAPE (%)	0.33	0.11	0.25	0.12
EM MAPE (%)	2.65	3.18	2.71	3.71

Faced with different excitation functions, from simple to complicated, 3D CoSim has maintained high accuracy. Regarding EM simulation, the near-port simulation preserves extremely high accuracy. Although the accuracy at the far port slightly declines, it stays within an acceptable range. As far as thermal field simulation is concerned, 3D CoSim offers high-accuracy predictions.

F. Speed Up

Comsol is a multi-physics field simulation software known for its high accuracy and performance. Here we mainly compare the speedup of 3D CoSim with that of Comsol. In the test case on Chip 1, Comsol takes 152 seconds to simulate an AMD EPYC 9754 128-core Processor CPU, while 3D CoSim prediction time is 0.18s on the same CPU and 0.02s on the Nvidia A100 GPU, which achieves 844× and 7600× speedup, respectively. It is worth noting that as excitation functions and chip structures grow in complexity, tools relying on conventional methods encounter escalating computational demands. In contrast, 3D CoSim, harnessing the power of operator learning, offers remarkable speed advantages, consistently delivering efficient solutions.

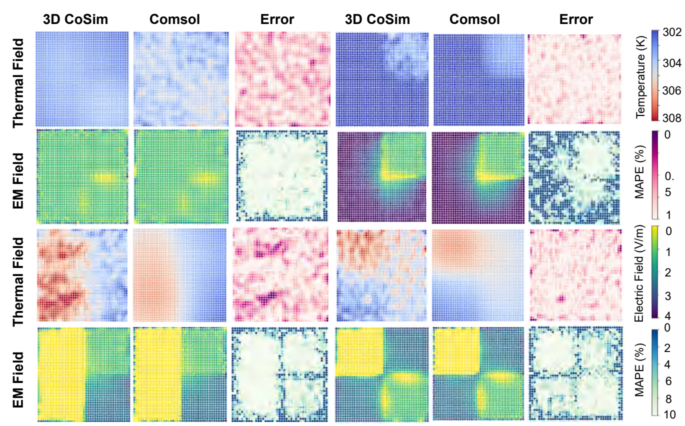

Fig. 7: Physical fields at the predicted excitation port interface (z=0) for different excitation functions.

G. Comparisons to SOTA Baselines

The application of machine learning methods in the analysis of physical fields has witnessed a dramatic increase recently. Although the existing works [14-17] address the co-simulation problem, they are not fully AI-based simulation methods and have disadvantages in predicting cost. Given the limited prior work on the coupled prediction of electromagnetic (EM) and thermal fields, we evaluate the performance of 3D CoSim by

TABLE IV: The Comparisons between Baseline models and 3D CoSim.

Baseline	[19] ANN	[20] NeuroTF	[21] ComplexTF	[21] DeepOTF	[22] Complex-DeepONet	3D CoSim (Ours)
Average error (%)	7.75	88.67	4.32	3.05	3.01	**3.00**
Runtime (s)	54	0.3	-	0.29	0.10	**0.02**
Field	EM	EM	EM	EM	EM	**EM & Thermal**
Baseline	[23] DeepOHeat	[24] ARO	[25] ThermTransformer	[26] ThermGAN	[27] LSTM	3D CoSim (Ours)
Average error (%)	0.08	**0.023**	0.36	0.60	2.19	0.20
Runtime (s)	**0.001**	0.23	0.014	0.016	0.019	0.02
Field	Thermal	Thermal	Thermal	Thermal	Thermal	**EM & Thermal**

comparing it with state-of-the-art (SOTA) baselines that are each tailored for single-physics field prediction. This comparison provides a holistic evaluation of 3D CoSim, assessing not only its capability to handle multi-physics fields but also its performance in predicting individual physics fields. The comparison results are demonstrated in the Table IV.

In terms of EM fields, we compare 3D CoSim with the following strong baselines:(1) ANN Predictor with Single MLP [19]; (2) NeuroTF with Pole-Residue-Based transfer functions [20]; (3) ComplexTF with Complex-Neural-Network-Based transfer function [21]; (4) DeepOTF with Data-Driven-Based transfer function [21]; (5) Complex-DeepONet with Operator-Learning-Based Field Predictor [22]. In the aforementioned cases, Baselines (1)–(4) focus primarily on predicting the transfer function of the EM field. These models are relatively simpler in structure and are designed to predict parameters S/Y/Z rather than the entire field distribution. Specifically, they do not predict the EM field values at all points within the domain, nor do they provide detailed insights into the field distribution concerning the underlying geometrical configurations and physical characteristics. As a result, these models exhibit certain limitations in capturing the comprehensive behavior of the EM field. As for baseline (5), it is designed to be predictive for specific structures but cannot generalize to different geometric configurations and excitation functions. This limitation restricts its applicability in scenarios where diverse geometries or varying excitation functions are encountered. Numerically, 3D CoSim demonstrates substantial advantages over other methods. It not only demonstrates the smallest prediction error among all the compared methods, with an evaluation error as low as 3.00%, but also boasts the fastest prediction speed at 0.02 seconds on the GPU.

To evaluate the progress achieved by the proposed 3D CoSim model in thermal field prediction, we selected five sets of strong baselines for comparison likewise: (1) Physics-Informed-Operate-Learning-Based DeepOHeat [23]; (2) Autoregressive-Operator-Learning-Based ARO [24]; (3) Self-Attention-Architecture-Based ThermTransformer [25]; (4) Generative-Adversarial-Learning-Based ThermGAN [27]; (5) Temporal-Aware Long-Short-Term-Memory-Based RealMaps [26]. It is important to note that while some single-physics baselines surpass 3D CoSim in thermal prediction accuracy, all these methods are limited to modeling the thermal field alone and do not account for its coupling with the EM field. As a result, the Joule heating effect induced by EM is not considered. This omission leads to an incomplete simulation that fails to capture the full complexity of the coupled physical phenomena. The prediction accuracy and running speed of 3D CoSim, which is capable of coupled-field prediction, are remarkably close to those of the current SOTA baselines. Specifically, 3D CoSim attains an average prediction error of only 0.20%.

Overall, 3D CoSim demonstrates exceptional performance in both EM and thermal field prediction, achieving high accuracy and remarkable computational speed. Moreover, the outstanding generalization ability of 3D CoSim enables it to effectively manage 3D ICs under a wide range of geometric configurations and excitation functions. This versatility is crucial for addressing the diverse and intricate challenges posed by modern 3D IC designs, making 3D CoSim a highly adaptable tool in the field of coupled-field analysis.

V. Conclusion

In this work, *for the first time*, we propose a multiphysics field coupling simulator 3D CoSim, realizing ultra-fast and high-precision EM-Thermal field predictions. Based on a tightly coupled operator learning architecture, we can learn to map nonlinear functions from geometric and physical characteristics to complex multiphysics fields. Coupled-field models are sequentially trained with data-driven and physics-informed loss by encoding geometrical constructions and physical setups under different physical fields. Experiments show that a trained 3D CoSim can predict with high accuracy the coupling fields of 3D ICs with unseen geometric configurations and excitation functions. 3D CoSim achieves an impressive 7600× speedup over traditional COMSOL solvers, while maintaining 97% accuracy for EM simulation and 99.8% for thermal simulation. Future work will focus on extending the capabilities of 3D CoSim to accommodate more intricate geometrical structures and industrial-scale chip layouts.

Acknowledgment

This work was financially supported by Project of Shanghai EC (24KXZNA12). The computations in this research were performed using the CFFF platform of Fudan University.

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